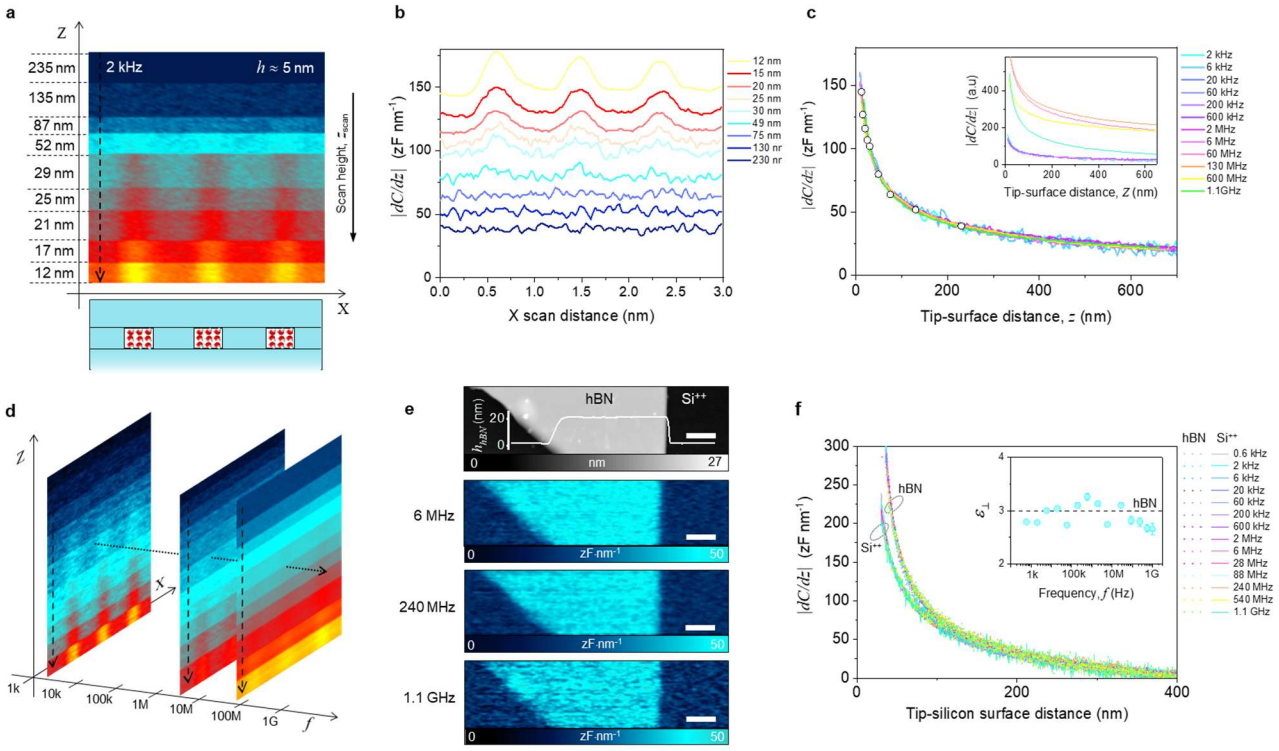
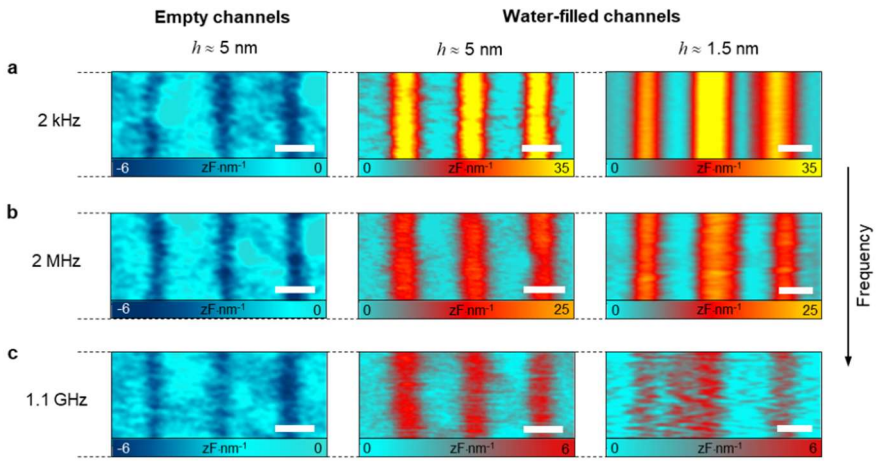
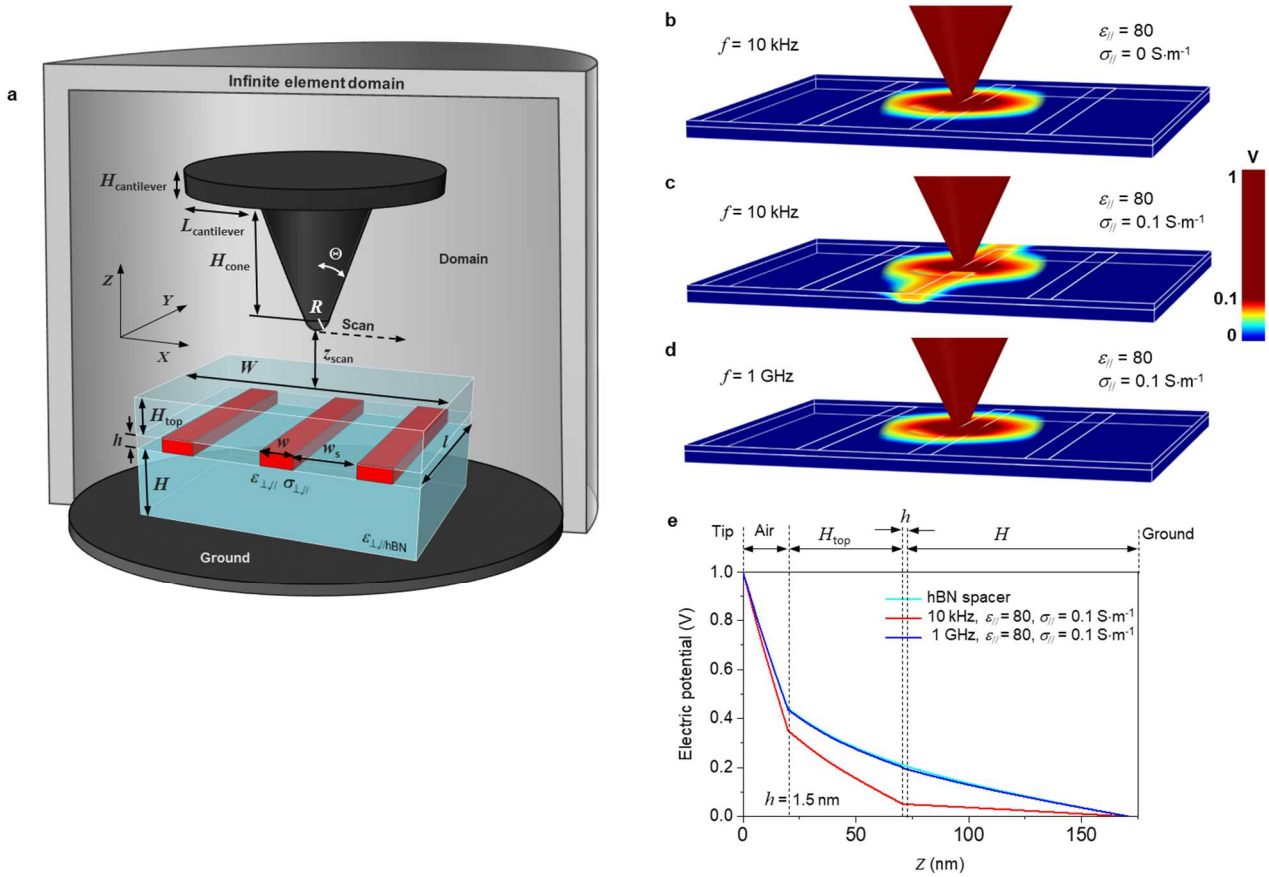


Extended Data Fig. 3 | Measurement and calibration procedures for local broadband dielectric spectroscopy. (a) Representative $|dC/dz|(x,z)$ image taken at 2 kHz for channels with $h \approx 5$ nm. The image shows 3 nanochannels and was acquired by measuring $|dC/dz|$ at various constant heights and a fixed y position. The tip approached the surface from above using small steps. **(b)** $|dC/dz|$ for scanning along the lateral (x) axis at constant y and various heights. The profiles were averaged over ~10 lines using images in (a). The signal became sensitive to the presence of water inside the channels at $z \lesssim$ 30 nm. **(c)** Gain-calibrated $|dC/dz|(z)$ curves at different frequencies taken over the center of the spacer region (color-coded lines). White symbols indicate the values of $|dC/dz|$ and z_{scan} corresponding to the profiles shown in (b). Inset: Same curves without calibration of the electrical gain g and the offsets (color-coded). We used the curve at 2 kHz as a reference to calibrate g at high frequencies by matching the reference curve with the curves taken at higher f, using only g and an offset as fitting parameters. **(d)** Schematic representation of our 3D AFM data sets $|dC/dz|(x,z,f)$ obtained by acquiring subsequent images $|dC/dz|(x,z)$ for many frequencies from kHz to GHz. The black dashed arrows in (a) and (d) indicate the position where the $|dC/dz|(z)$ curves in (c) were acquired before taking such 3D images. **(e,f)** Example of broadband dielectric imaging spectroscopy on a test sample of hBN on doped silicon using the same g calibration procedure: topography image (grey colored) of 21 nm-thick hBN crystal on doped Si; and corresponding calibrated dielectric images (cyan colored images) at frequencies 6 MHz, 240 MHz and 1.1 GHz at a constant height (~45 nm). Scale bar, 1 μm. The curves in (f) are gain-calibrated $|dC/dz|(z)$ curves taken above the Si substrate (solid curves) and the hBN layer (dotted) at various f. The gain g for each f was obtained by scaling the curves measured on doped Si to the low-f reference curve at 2 kHz using only g and an offset as fitting parameters. By fitting the calibrated curves on hBN using the procedure and analytical formula in Ref.[65], we obtained the dielectric constant of the hBN layer as function of f (inset). Its average value across all the f up to GHz was found to be $\varepsilon_{\perp hBN} = 2.9 \pm 0.2$, which is in good agreement with the expected bulk value for hBN in the perpendicular direction ($\varepsilon_{\perp hBN} \approx 3$, dashed line). Bars: standard error defined as 95% confidence interval of the fitting, not shown when smaller than the symbol. Other fitting parameters: AFM tip's R = 102 nm and θ = 25°, obtained by fitting the curve taken on doped Si at 2 kHz.



Extended Data Fig. 4| Additional dielectric images. (**a-c**) Dielectric images of three channels taken at constant height and at characteristic f of 2 kHz, 2 MHz and 1.1 GHz, respectively. Same devices as in Fig. 2b of the main text: $h \approx 5$ and 1.5 nm, empty and filled with water (as specified in the legends). Scalebars, 500 nm.



Extended Data Fig. 5 | 3D numerical simulations. (**a**) Schematics (not to scale) of the 3D numerical model used to fit the experimental data. (**b**-**d**) Calculated electrostatic potential for h = 1.5 nm and three representative cases: (b) low (10 kHz) frequency and purely dielectric channels ($\varepsilon_{//}$ = 80 and $\sigma_{//}$ = 0); (c) Same low frequency but conductive channels ($\varepsilon_{//}$ = 80 and $\sigma_{//}$ = 0.1 S·m^{-1}); (d) same as (c) but at f beyond conductivity relaxation. (**e**) Potential along the Z direction (counted from the AFM tip) for the tip placed at z_{scan} = 20 nm above the middle of the water channel (red and blue) and above the center of the spacer region (cyan). Frequencies 10 kHz and 1 GHz, color coded. Water channel's parameters: $\varepsilon_{//}$ = 80, $\sigma_{//}$ = 0.1 S·m^{-1}, $\varepsilon_{\perp}$ = 2, $\sigma_{\perp}$ = 0, l = 3 µm, w_c = 200 nm; w_s = 800 nm. Other parameters: R = 100 nm, θ = 25°, H_{cone} = 6 µm, $H_{cantilever}$ = 3 µm, $L_{cantilever}$ = 0 µm, H_{top} = 50 nm, H = 100 nm, $\varepsilon_{//hBN}$ = 5, $\varepsilon_{\perp hBN}$ = 3.